

DataBridge

Technical Architecture Overview

AI-Powered Semantic Data Platform

Version 1.0 | April 2026

Confidential | EpicData BV

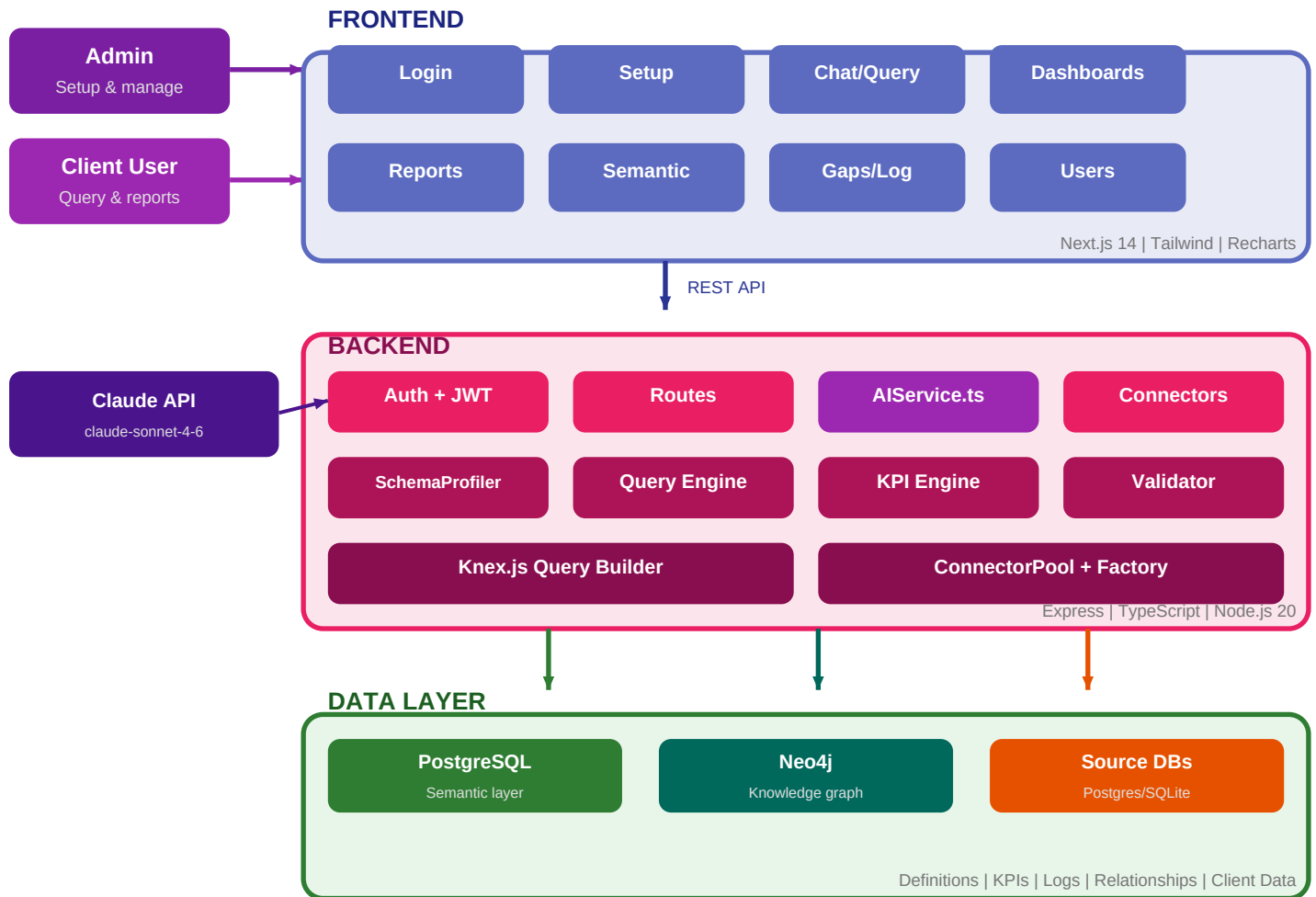
Architecture & Infrastructure Documentation

Azure Container Apps | PostgreSQL | Neo4j | Claude AI

Next.js | Express | TypeScript | Terraform

1. System Overview

DataBridge enables business users to query their data using natural language, powered by a semantic layer and Claude AI.



Key Components:

Frontend: 8 pages — login, setup wizard, chat query, dashboards, reports, semantic definitions, gaps/logs, user management

Backend: Express API with JWT auth, AI service, connector pool, schema profiler, query engine with repair loop

Data: PostgreSQL stores metadata + semantic layer. Neo4j stores knowledge graph. Source DBs hold client business data.

AI: All Claude calls route through AIService.ts — schema drafts, NL-to-SQL, answer formatting, dashboard specs, report narratives

2. Azure Infrastructure

All services run in Azure West Europe (Netherlands). Container Apps scale to zero when idle for cost optimization.

Resource Group: databridge-prod-rg

Container Apps Environment

PUBLIC (HTTPS)

Frontend

Next.js | 0.25 CPU

SCALE-TO-0

Backend

Express | 0.5 CPU

SCALE-TO-0

PostgreSQL

Flexible Server B1ms

ALWAYS ON

~25 EUR/mo

INTERNAL ONLY

Neo4j

Graph DB | 0.5 CPU

SCALE-TO-0

ETL Worker

Python | 0.5 CPU

SCALE-TO-0

Azure File Share

Neo4j persistent data

Key Vault

Secrets (JWT, API keys)

Container Registry

Docker images (Basic)

Blob Storage

Warehouse + files

Application Insights

Monitoring & logs

Internet

HTTPS / Firewall

Users access Frontend + Backend via HTTPS

Cost Summary

Idle (no traffic): ~30 EUR/month

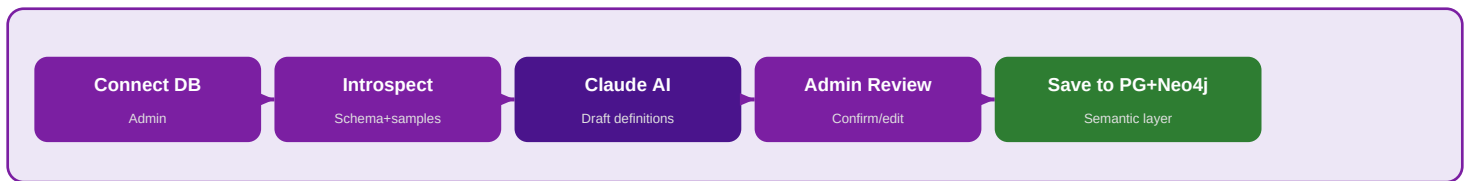
Active (with usage): ~50-60 EUR/month

PostgreSQL is the main cost (~25 EUR/mo)

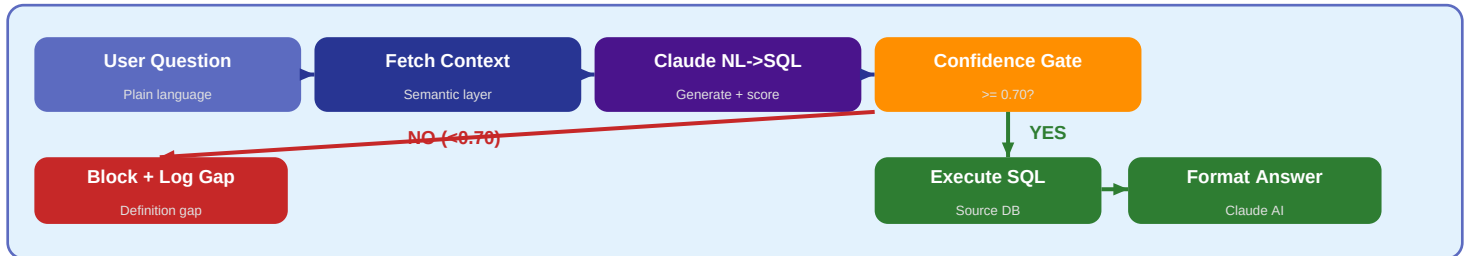
Container Apps cost 0 EUR when scaled to zero

3. Data Flow Architecture

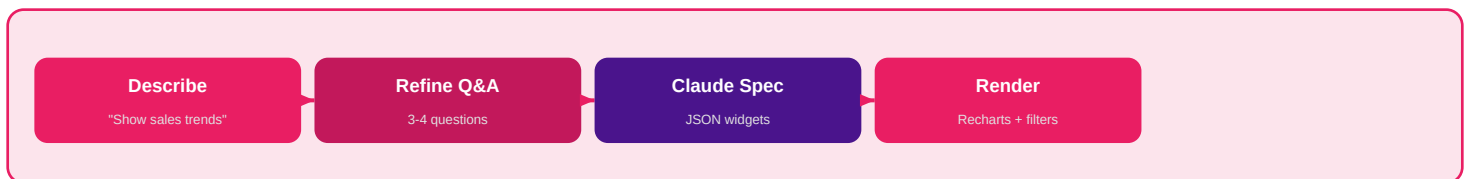
Flow 1: Setup & Schema Profiling



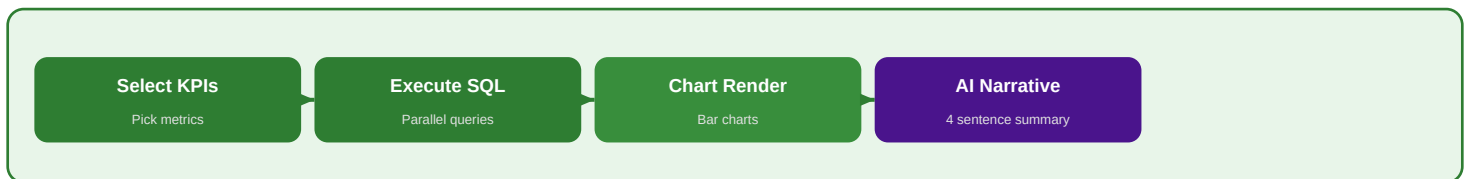
Flow 2: Natural Language Query



Flow 3: AI Dashboard Generation

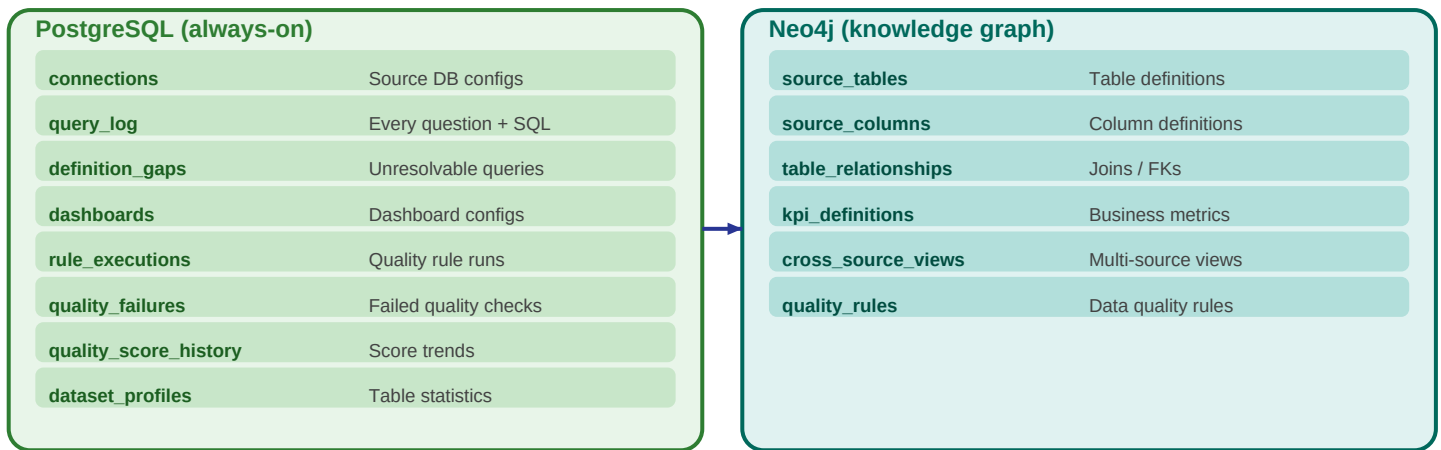


Flow 4: KPI Report Builder

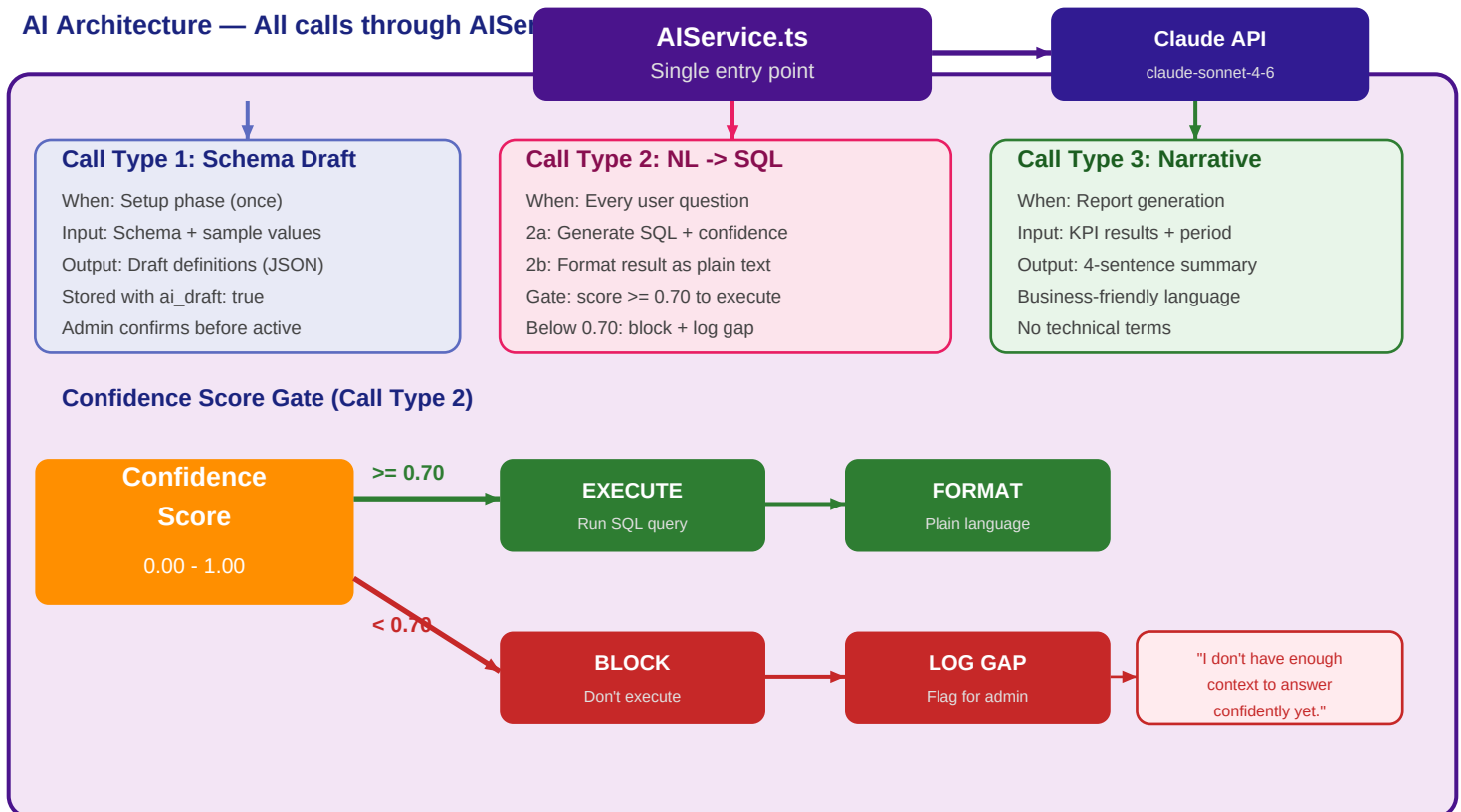


4. Semantic Layer & AI Architecture

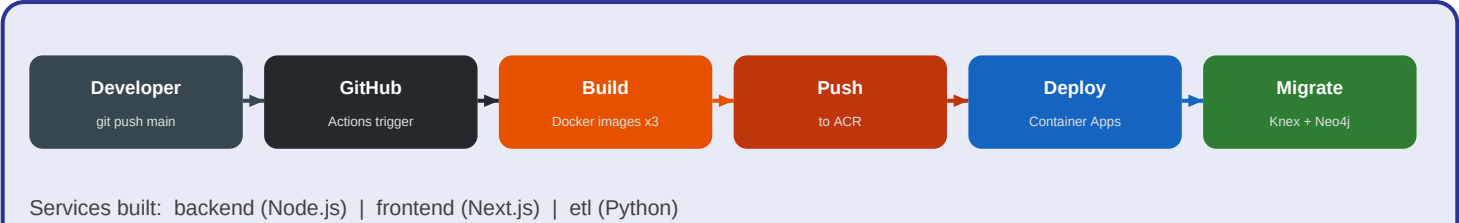
Semantic Layer Data Model



AI Architecture — All calls through AIService.ts



5. CI/CD Pipeline



6. Tech Stack

Category	Technology	Purpose
Frontend	Next.js 14, Tailwind CSS	App Router, SSR, responsive UI
	Recharts	Interactive charts & dashboards
	Axios	API client with auth interceptor
Backend	Express.js + TypeScript	REST API, middleware, routes
	Knex.js	Query builder for PG + SQLite
	@anthropic-ai/sdk	Claude API integration
Databases	PostgreSQL 16	Semantic layer, metadata, logs
	Neo4j 5 Community	Knowledge graph (relationships)
	Source: Postgres/SQLite	Client business data (read-only)
AI	Claude claude-sonnet-4-6	NL-SQL, schema drafts, narratives
Infrastructure	Azure Container Apps	Scale-to-zero containers
	Terraform	Infrastructure as Code

7. Security

Authentication	JWT tokens with 8h expiry, bcrypt password hashing, role-based access (admin/client)
Encryption	AES-256 encryption for database credentials at rest, SSL/TLS for all connections
Secrets	Azure Key Vault stores JWT secret, Anthropic API key, encryption key — never in code
Network	Neo4j + ETL are internal-only (no public URL). Postgres firewall restricts IP access
Data Safety	Source databases are read-only. AI never modifies client data. SQL validated before execution
Monitoring	Application Insights tracks errors, response times. Query log captures every AI-generated SQL

8. Cost Breakdown & Roles

Monthly Cost Breakdown (EUR)

Component	Idle Cost	Active Cost	Notes
PostgreSQL Flexible Server	~25	~25	B1ms (1 vCPU, 2GB) — always on
Container Apps: Backend	0	~8-12	0.5 CPU, 1GB — scale-to-zero
Container Apps: Frontend	0	~3-5	0.25 CPU, 0.5GB — scale-to-zero
Container Apps: Neo4j	0	~5-8	0.5 CPU, 1GB — scale-to-zero
Container Apps: ETL	0	~3-5	0.5 CPU, 1GB — scale-to-zero
Container Registry (ACR)	~5	~5	Basic tier — stores Docker images
Blob Storage	<1	~1-2	LRS — warehouse + Neo4j file share
Key Vault	0	<1	Standard tier — low usage free
Application Insights	0	~1-2	First 5GB/month free
TOTAL	~30	~50-60	

Roles & Permissions

Feature	Admin	Client User
Connect data source	Y	-
Run schema introspection	Y	-
Review / confirm definitions	Y	-
Add / edit KPI definitions	Y	-
View definition gaps	Y	-
Ask questions (chat)	Y	Y
Build and view reports & dashboards	Y	Y
View full query log	Y	-
See SQL toggle (show query)	Y	-
Manage users	Y	-

Deployment

Frontend: <https://databridge-prod-frontend.happysand-8549b35c.westeurope.azurecontainerapps.io>

Backend: <https://databridge-prod-backend.happysand-8549b35c.westeurope.azurecontainerapps.io>